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EDEXCEL INTERNATIONAL GCSE MATH (4MA1) • UNIT 3.01

Functions, Transformations & Arithmetic Series

ANSWER KEY & SOLUTIONS

25 Structured Exam-Style Questions

Total Questions: 25 • Verified Step-by-Step Solutions

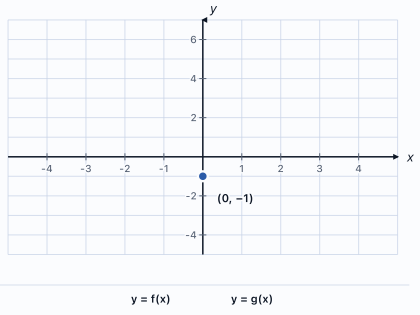
QUICK-REFERENCE ANSWER SUMMARY (25 ITEMS)

Q1 C	Q2 D	Q3 A	Q4 B	Q5 C
Q6 D	Q7 A	Q8 B	Q9 C	Q10 D
Q11 D	Q12 A	Q13 A	Q14 C	Q15 B
Q16 B	Q17 B	Q18 A	Q19 B	Q20 B
Q21 D	Q22 A	Q23 C	Q24 C	Q25 A

Functions, Transformations & Arithmetic Series Practice Exercises

1 MCQ

The coordinate diagram shows the parabolic curve $f_1(x) = x^2 - 10x + 38$ in the domain of Functions, Transformations & Arithmetic Series. By completing the square $f_1(x) = (x-h)^2 + k$, find the sum $k + f_1(7)$.



A 35

B 17

C 30 ✓

D 28

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Complete the square: $x^2 - 10x + 38 = (x-5)^2 - 25 + 38 = (x-5)^2 + 13$, giving minimum value $k = 13$. **[M1]**
2. Evaluate at $x = 7$: $f_1(7) = (2)^2 + 13 = 17$. **[M1]**
3. Sum the two quantities: $k + f_1(7) = 13 + 17 = 30$. **[A1]**

- **Final Verified Answer:** 30.

2 MCQ

A mathematical model in Functions, Transformations & Arithmetic Series is defined by the cubic function $y_2(x) = 4x^3 - 5x^2 + 15x - 10$. Calculate the exact gradient of the tangent to the curve at the point where $x = 3$.

A 123

B 78

C 41

D 93 ✓

WORKED SOLUTION & VERIFICATION

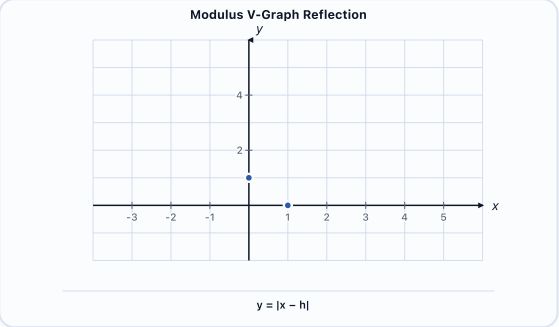
Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 10x + 15$. **[M1]**
2. Substitute $x = 3$: $\left.\frac{dy}{dx}\right|_{x=3} = 12(9) - 10(3) + 15 = 93$. **[M1][A1]**

- **Final Verified Answer:** 93.

3 MCQ

In Functions, Transformations & Arithmetic Series, two displacement vectors are given by $\mathbf{u}_3 = \langle 5, -8 \rangle$ and $\mathbf{v}_3 = \langle 9, 7 \rangle$. Compute the exact scalar value of $\mathbf{u}_3 \cdot \mathbf{v}_3 + \|\mathbf{u}_3\|^2$.



A 78 ✓

B -11

C 89

D 190

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Compute the inner product: $\mathbf{u}_3 \cdot \mathbf{v}_3 = (5)(9) + (-8)(7) = 45 + (-56) = -11$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_3\|^2 = 5^2 + (-8)^2 = 89$. **[M1]**
3. Combine: $-11 + 89 = 78$. **[A1]**

- **Final Verified Answer:** 78.

4 MCQ

The linear growth profile shown in the diagram for Functions, Transformations & Arithmetic Series follows an arithmetic progression with first term $a_1 = 6$ and common difference $d = 6$. Find the exact sum S_{15} of the first 15 terms.

A 90

B 720 ✓

C 810

D 1350

WORKED SOLUTION & VERIFICATION

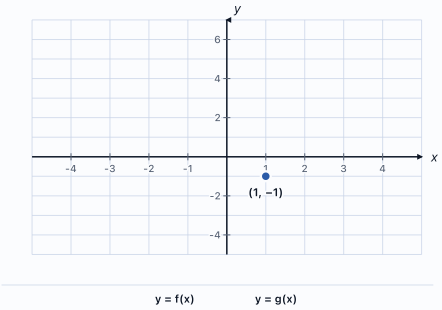
Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Use the sum formula for an arithmetic series: $S_n = \frac{n}{2}[2a_1 + (n-1)d]$. **[M1]**
2. Substitute $n = 15, a_1 = 6, d = 6$: $S_{15} = \frac{15}{2}[12 + 14 \times 6] = \frac{15}{2} \times 96 = 720$. **[M1][A1]**

- **Final Verified Answer:** 720.

5 MCQ

In Functions, Transformations & Arithmetic Series, evaluate the exact definite integral $I_5 = \int_0^2 (18x^2 + 20x)\,\mathrm{d}x$.



A 112

B 108

C 88 ✓

D 45

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Integrate term by term: $\int (18x^2 + 20x)\,\mathrm{d}x = [6x^3 + 10x^2]_0^2$. **[M1]**
2. Substitute the upper and lower limits: $(6(2)^3 + 10(2)^2) - 0 = 48 + 40 = 88$. **[M1][A1]**

- **Final Verified Answer:** 88.

6 MCQ

In Functions, Transformations & Arithmetic Series, consider the quadratic function $f_6(x) = x^2 - 12x + 121$. Express $f_6(x)$ in the completed-square form $(x-h)^2 + k$ and compute the exact value of $k + f_6(13)$.

A 225

B 134

C 212

D 219 ✓

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Complete the square: $x^2 - 12x + 121 = (x-6)^2 - 36 + 121 = (x-6)^2 + 85$, giving minimum value $k = 85$. **[M1]**
2. Evaluate at $x = 13$: $f_6(13) = (7)^2 + 85 = 134$. **[M1]**
3. Sum the two quantities: $k + f_6(13) = 85 + 134 = 219$. **[A1]**

- **Final Verified Answer:** 219.

7 MCQ

The graph illustrates the cubic model $y_7(x) = 4x^3 - 6x^2 + 33x - 17$ studied in Functions, Transformations & Arithmetic Series. Find the instantaneous gradient $\frac{dy}{dx}$ of the tangent to the curve at $x = 8$.

A 705 ✓

B 801

C 672

D 117

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Differentiate term by term using the power rule: $\frac{dy}{dx} = 12x^2 - 12x + 33$. **[M1]**
2. Substitute $x = 8$: $\left.\frac{dy}{dx}\right|_{x=8} = 12(64) - 12(8) + 33 = 705$. **[M1][A1]**

- **Final Verified Answer:** 705.

8 MCQ

In Functions, Transformations & Arithmetic Series, two displacement vectors are given by $\mathbf{u}_8 = \langle 6, -7 \rangle$ and $\mathbf{v}_8 = \langle 8, 5 \rangle$. Compute the exact scalar value of $\mathbf{u}_8 \cdot \mathbf{v}_8 + \|\mathbf{u}_8\|^2$.

A 13

B 98 ✓

C 85

D 168

WORKED SOLUTION & VERIFICATION

Mark Scheme & Examiner Solution (Functions, Transformations & Arithmetic Series):

1. Compute the inner product: $\mathbf{u}_8 \cdot \mathbf{v}_8 = (6)(8) + (-7)(5) = 48 + (-35) = 13$. **[M1]**
2. Compute the squared norm: $\|\mathbf{u}_8\|^2 = 6^2 + (-7)^2 = 85$. **[M1]**
3. Combine: $13 + 85 = 98$. **[A1]**

- **Final Verified Answer:** 98.

9 MCQ

An arithmetic series has first term $a = 4$ and common difference $d = 3$. Calculate the sum S_{12} of the first 12 terms.

A 282

B 444

C 246 ✓

D 37

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{12} = \frac{12}{2}[2(4) + 11(3)] = 246$.

10 MCQ

An arithmetic series has first term $a = 5$ and common difference $d = 4$. Calculate the sum S_{12} of the first 12 terms.

A 372

B 588

C 49

D 324 ✓

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{12} = \frac{12}{2}[2(5) + 11(4)] = 324$.

11 MCQ

An arithmetic series has first term $a = 6$ and common difference $d = 5$. Calculate the sum S_{12} of the first 12 terms.

A 732

B 462

C 61

D 402 ✓

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{12} = \frac{12}{2}[2(6) + 11(5)] = 402$.

12 MCQ

An arithmetic series has first term $a = 7$ and common difference $d = 6$. Calculate the sum S_{12} of the first 12 terms.

A 480 ✓

B 73

C 552

D 876

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{12} = \frac{12}{2}[2(7) + 11(6)] = 480$.

13 MCQ

An arithmetic series has first term $a = 8$ and common difference $d = 7$. Calculate the sum S_{13} of the first 13 terms.

- **A** 650 ✓
- **B** 1196
- **C** 741
- **D** 92

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{13} = \frac{13}{2}[2(8) + 12(7)] = 650$.

14 MCQ

An arithmetic series has first term $a = 9$ and common difference $d = 8$. Calculate the sum S_{13} of the first 13 terms.

- **A** 105
- **B** 1365
- **C** 741 ✓
- **D** 845

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{13} = \frac{13}{2}[2(9) + 12(8)] = 741$.

15 MCQ

An arithmetic series has first term $a = 10$ and common difference $d = 3$. Calculate the sum S_{13} of the first 13 terms.

- **A** 403
- **B** 364 ✓
- **C** 46
- **D** 598

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{13} = \frac{13}{2}[2(10) + 12(3)] = 364$.

16 MCQ

An arithmetic series has first term $a = 11$ and common difference $d = 4$. Calculate the sum S_{13} of the first 13 terms.

- **A** 767
- **B** 455 ✓
- **C** 507
- **D** 59

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{13} = \frac{13}{2}[2(11) + 12(4)] = 455$.

17 MCQ

An arithmetic series has first term $a = 12$ and common difference $d = 5$. Calculate the sum S_{14} of the first 14 terms.

A 1078

B 623 ✓

C 77

D 693

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{14} = \frac{14}{2}[2(12) + 13(5)] = 623$.

18 MCQ

An arithmetic series has first term $a = 13$ and common difference $d = 6$. Calculate the sum S_{14} of the first 14 terms.

A 728 ✓

B 1274

C 812

D 91

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{14} = \frac{14}{2}[2(13) + 13(6)] = 728$.

19 MCQ

An arithmetic series has first term $a = 14$ and common difference $d = 7$. Calculate the sum S_{14} of the first 14 terms.

A 1470

B 833 ✓

C 931

D 105

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{14} = \frac{14}{2}[2(14) + 13(7)] = 833$.

20 MCQ

An arithmetic series has first term $a = 4$ and common difference $d = 8$. Calculate the sum S_{14} of the first 14 terms.

A 108

B 784 ✓

C 896

D 1512

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{14} = \frac{14}{2}[2(4) + 13(8)] = 784$.

21 MCQ

An arithmetic series has first term $a = 5$ and common difference $d = 3$. Calculate the sum S_{15} of the first 15 terms.

A 705	B 47
C 435	D 390 ✓

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{15} = \frac{15}{2}[2(5) + 14(3)] = 390$.

22 MCQ

An arithmetic series has first term $a = 6$ and common difference $d = 4$. Calculate the sum S_{15} of the first 15 terms.

A 510 ✓	B 930
C 62	D 570

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{15} = \frac{15}{2}[2(6) + 14(4)] = 510$.

23 MCQ

An arithmetic series has first term $a = 7$ and common difference $d = 5$. Calculate the sum S_{15} of the first 15 terms.

A 1155	B 77
C 630 ✓	D 705

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{15} = \frac{15}{2}[2(7) + 14(5)] = 630$.

24 MCQ

An arithmetic series has first term $a = 8$ and common difference $d = 6$. Calculate the sum S_{15} of the first 15 terms.

A 92	B 1380
C 750 ✓	D 840

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{15} = \frac{15}{2}[2(8) + 14(6)] = 750$.

25 MCQ

An arithmetic series has first term $a = 9$ and common difference $d = 7$. Calculate the sum S_{16} of the first 16 terms.

- **A** 984 ✓
- **B** 1824
- **C** 1096
- **D** 114

WORKED SOLUTION & VERIFICATION

Using $S_n = \frac{n}{2}[2a + (n-1)d]$: $S_{16} = \frac{16}{2}[2(9) + 15(7)] = 984$.